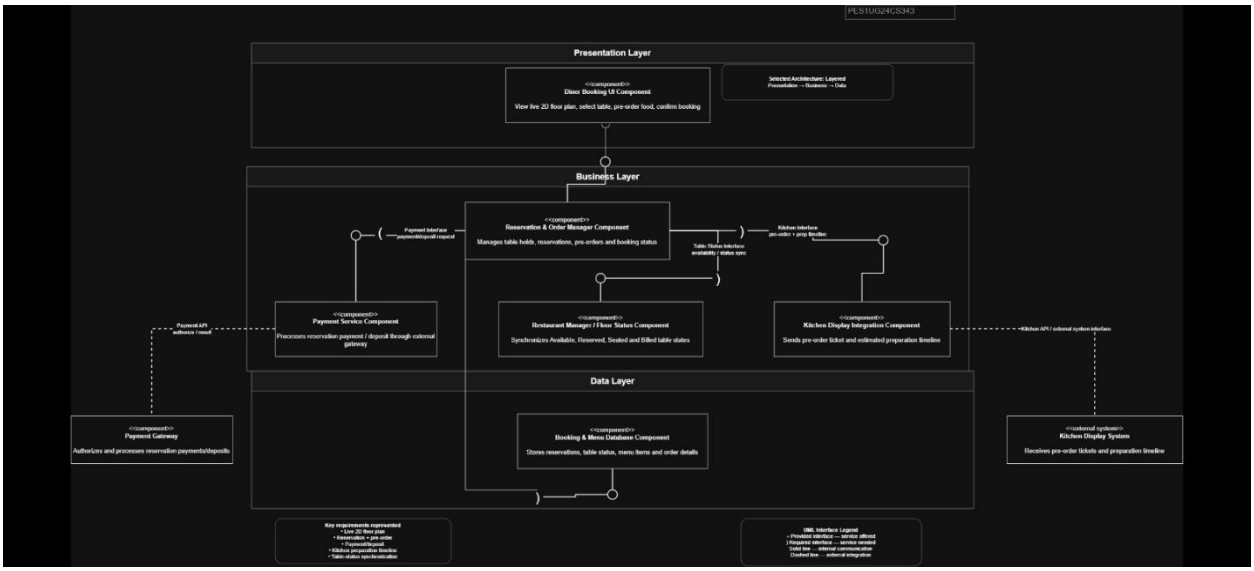
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SECTION:	F

SOFTWARE ENGINEERING

LAB 3

Deliverable 1 UML diagram



Deliverable 2:

Problem Statement #36 – Restaurant Table Booking & Pre-Ordering App

Architecture Selection: Layered Architecture

We chose Layered Architecture for the Restaurant Table Booking & Pre-Ordering App. The system is organized into three layers: Presentation, Business, and Data. The Presentation Layer handles diner interactions through the booking interface, the Business Layer manages reservations, pre-orders, payments, table status, and kitchen integration, while the Data Layer stores reservation, menu, table-status, and order information. This follows the layered architecture structure described in the lab, which separates the system into horizontal layers with distinct responsibilities.

Architectural Choice and Reasons

1. Clear separation of responsibilities:

Layered Architecture separates the user interface from the core business operations and data storage. The Diner Booking UI can handle viewing the live 2D floor plan, selecting an available table, adding pre-order items, and confirming the booking, while the Reservation & Order Manager handles the reservation and pre-order workflow. Payment, table-status synchronization, and kitchen communication are handled by separate business components.

2. Simplicity and maintainability:

The system has clearly defined functions such as table reservation, food pre-ordering, payment/deposit processing, cancellation, and table-status management. A layered structure keeps these responsibilities organized and makes the system easier to understand, maintain, and modify. It also avoids the additional operational complexity that would be introduced by using multiple independently deployed microservices for this application.

Security Advantage

Layered Architecture provides separation between the presentation, business, and data layers. The Diner Booking UI does not directly access the database or external payment gateway. Requests are handled through the business layer, allowing access to reservations, table status, menu data, and payment processing to be controlled through appropriate components. This reduces direct exposure of application data and supports secure payment processing through the Payment Service and Payment Gateway.

Performance Benefit

The layered architecture allows frequently used booking operations to be handled within the application's business layer without unnecessary distributed communication between many independent services. This helps reduce communication overhead and supports responsive operations such as viewing table availability, creating reservations, and processing pre-orders. The architecture can also support the requirement that table-status synchronization across manager terminals occur within 500 ms